

# Jacob Daniel R

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## Professional Summary

Data Engineer focused on building end-to-end data pipelines and cloud platforms. Skilled in SQL, Python, PySpark, and data warehousing, with hands-on experience in AWS, Azure Databricks, Snowflake, and Airflow. Worked on ETL, data transformation, validation, and workflow automation for analytics and AI/ML.

## Key Skills

**Programming:** SQL, Python (Numpy, Pandas), PySpark

**Data Engineering:** ETL/ELT Pipelines, Incremental Loading, Data quality/validation, Data Pipelines, Data Integration, Data Processing, Lakehouse Architecture, Medallion Architecture

**Cloud:** Azure Databricks, AWS (S3, Glue, Lambda, DynamoDB), ADLS Gen2, Unity Catalog, IAM

**Databases & Modeling:** Snowflake, Databricks, Redshift, MySQL, Data Warehousing, Data Modeling, Dimensional Modeling, Delta Lake, Data Pipeline Orchestration

**Tools:** Apache Airflow, Git/GitHub, Docker, Power BI

## Professional Experience

### AtliQ Technologies

Remote

Data Engineer (Internship)

Jun 2026 - Aug 2026

- Created a serverless AWS ETL pipeline using S3, Glue, Lambda, DynamoDB and PySpark to validate 1,000+ Finance and Sales records.
- Developed a Silver-layer workflow in Azure Databricks and Delta Lake, standardizing 3 data attributes and adding business metrics to datasets.
- Consolidated 3 Lambda functions into 1 ingestion flow supporting 4 regions: Asia, EMEA, North America and LATAM.
- Applied 4+ data quality rules with schema-drift handling, achieving 97–98% data quality across 500–1,000 records per file.
- Centralized audit tracking in 1 DynamoDB table with Slack alerts and least-privilege IAM.

## Projects

### Automated EOD Securities Pricing Platform - Securities Market Analytics [Project Link](#)

[Apache Airflow / Snowflake / AWS / Python / SQL]

Jul 2026 - Aug 2026

- Streamlined U.S. securities EOD data processing with 2 ingestion modes: historical backfill and daily incremental loads.
- Orchestrated 8 ordered Snowflake tasks through Apache Airflow for ingestion, validation, transformation and monitoring.
- Structured the warehouse using 3 dimensions and 1 daily pricing fact with incremental MERGE processing.
- Formed 6 curated Snowflake views covering equity performance, returns, sector and ETF liquidity.
- Added a 10-stock watchlist and Top 20 volume analysis with Slack alerts and 3 Airflow retries.

### Azure Databricks Retail Lakehouse - Retail Analytics [Project Link](#)

[Azure Databricks / PySpark / Data Warehousing / ETL Pipelines / SQL]

May 2026 - Jun 2026

- Constructed a Medallion Lakehouse with 23 Delta tables across Bronze, Silver and Gold layers.
- Processed 1M+ retail records using Auto Loader, PySpark, Delta MERGE and incremental workflows.
- Developed 5 dimension and 3 fact pipelines with cleansing, validation, deduplication and business rules.
- Coordinated 4 Databricks workflows for daily and monthly data processing with dependency management.
- Enforced Unity Catalog permissions for governed access to analytics datasets.
- Published 8 analytics-ready tables with Unity Catalog governance for Power BI reporting.

### AWS Support Analytics Pipeline - Customer Support Analytics [Project Link](#)

[ETL Pipelines / Amazon S3 / AWS Lambda / AWS Glue / Amazon Redshift]

May 2026 - Jun 2026

- Devised an automatic AWS ETL pipeline combining 2 data sources: MySQL support tickets and daily backend logs.
- Handled 31 days of support data through S3, Lambda, Glue, Athena, Redshift and CloudWatch.
- Converted raw log files into Parquet using Regex parsing, duplicate removal and field standardization.
- Set up 2 incremental ingestion flows with S3 event triggers for Scheduled daily processing.
- Connected transformed data to Redshift Serverless using Triggered COPY operations and incremental warehouse loading.
- Reduced manual execution by ~2 hrs/week by automating transformation, warehouse loading and reporting.

## Certifications

- **Data Engineering Basics for Data Analysts - Codebasics**  
Issued on: Jun 2026 [Certification Link](#)

## Education

### Anna University

Bachelor of Engineering - Computer Science and Engineering  
CGPA: 8.38

Tamil Nadu, India  
Aug 2022 - May 2026